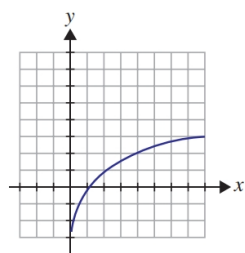


Logarithmic Functions

Section

3.2

A. Graph of a Logarithmic Function



A. Review

Expansion Properties for Logarithms

4. $\log_b(M \cdot N) = \log_b M + \log_b N$ (Product Rule)
5. $\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$ (Quotient Rule)
6. $\log_b(M^n) = n \cdot \log_b M$ (Power rule)

Examples

1. (video) For the function $f(x) = \frac{3x+1}{2x+4}$ the inverse function $f^{-1}(x)$ equals

2. (video) Expand the expression $\log\left(\frac{1000 x^4 \sqrt{z}}{(y+5)^2}\right)$ completely: log

3. (video) Write $4 \log x + \frac{1}{2} \log z - 2 \log(y + 3) + \log 8$ as a simple logarithm and simplify as much as possible.

B. Limit Rules

6. $\lim_{x \rightarrow 0^+} \ln x = -\infty$

7. $\lim_{x \rightarrow \infty} \ln x = \infty$

8. $\lim_{x \rightarrow 0^+} \log x = -\infty$

9. $\lim_{x \rightarrow \infty} \log x = \infty$

Examples

4. (video) Find $\lim_{x \rightarrow \infty} \frac{-2 \ln x + 3}{5}$

5. (video) Find $\lim_{x \rightarrow 0^+} \frac{\ln x + 3x - 2}{4x^2 - 3}$

6. $\lim_{x \rightarrow 3^+} \log(x^2 - 5x + 6) =$

7. $\lim_{x \rightarrow 0^+} \log(\sin x) =$

8. $\lim_{x \rightarrow \infty} [\log(1 + x^2) - \log(1 + x)] =$

9. If f is one-to-one and $f(3) = 11$, then

a. $f^{-1}(11) =$

b. $[f(3)]^{-1} =$

10. Find the inverse for each of the following:

a. $f(x) = \frac{4x - 12}{19x + 15}$

b. $h(x) = e^{9x+3}$

c. $f(x) = \ln(13x + 10)$

11. If $f(x) = x^3 + 1$

a. Find $f^{-1}(x) =$

b. Find $(f^{-1})'(x) =$

c. Find $(f^{-1})'(2) =$

C. "Derivative of Inverse" Formula

When we cannot find $f^{-1}(x)$ and $f(x)$ is one-to-one: $(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$

12. For $f(x) = x^3 + 4x + 4$, find $(f^{-1})'(4) =$

13. If $f(x) = \sqrt{x^3 + x^2 + x + 1}$ find $(f^{-1})'(1) =$

14. Suppose f^{-1} is the inverse function of a differentiable function f and $f(3) = 4$, $f'(3) = \frac{7}{4}$
then $(f^{-1})'(4) =$

15. If $\ln a = 2$, $\ln b = 3$, and $\ln c = 5$, evaluate $\ln(\sqrt{b^{-4}c^{-4}a^{-3}}) =$

16. Solve each equation for x :

a. $3^{x-4} = 9$

b. $\ln x + \ln(x-1) = 4$

